Practice workbook · Unit vectors: same direction, length one

Finding a vector of length one in the direction of a given vector by dividing the vector by its magnitude, including working backwards from a given unit vector to find unknown constants.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-03

1. Question 6 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Worked-example retry · Attempt part b only.

6 The points P, Q, R and S are the vertices of a quadrilateral $PQRS$ such that

$$\overrightarrow{PQ} = 2\mathbf{i} + 3\mathbf{j} \qquad \overrightarrow{PR} = -\mathbf{i} + 18\mathbf{j} \qquad \overrightarrow{PS} = -3\mathbf{i} + 15\mathbf{j}$$

(a) Show that $PQRS$ is a parallelogram.

(4)

(b) Find a unit vector parallel to $\overrightarrow{QS}$ as a simplified expression in terms of **i** and **j**

(4)

The point T lies on QS such that $QT : TS = 5 : 8$

(c) Find $\overrightarrow{PT}$ as a simplified expression in terms of **i** and **j**

(2)

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1. Question 6 · part b · 4 marks · Working and answer

Working and answer

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2. Question 7 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 O, A, B and C are fixed points such that

$$\overrightarrow{OA} = 8\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OB} = 15\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OC} = 8\mathbf{i} + \mathbf{j}$$

(a) Find $\overrightarrow{BC}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector parallel to $\overrightarrow{BC}$

(2)

The point M is the midpoint of OA and the point N lies on OB such that $ON : NB = 1 : 2$

(c) Show that the points M, N and C are collinear.

(4)

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2. Question 7 · part b · 2 marks · Working and answer

Working and answer

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3. Question 1 · part b · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

1 Referred to a fixed origin O, the point A has position vector $(4\mathbf{i} + 3\mathbf{j})$ and the point B has position vector $(\mathbf{i} + 7\mathbf{j})$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector that is parallel to $\overrightarrow{AB}$

(2)

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(Total for Question 1 is 4 marks)

3. Question 1 · part b · 2 marks · Working and answer

Working and answer

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4. Question 1 · part c · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

1 The position vector of the point A is $(3\mathbf{i} - 2\mathbf{j})$, referred to a fixed origin O.

The point B is such that $\overrightarrow{AB} = (6\mathbf{i} + 8\mathbf{j})$

(a) Find the position vector of B as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find the magnitude of vector $\overrightarrow{AB}$

(1)

(c) Find a unit vector, in terms of $\mathbf{i}$ and $\mathbf{j}$, that is parallel to $\overrightarrow{AB}$

(2)

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4. Question 1 · part c · 2 marks

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Further practice · Attempt part c only.

Question 1 continued

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(Total for Question 1 is 5 marks)

4. Question 1 · part c · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = 5\mathbf{i} + 7\mathbf{j} \qquad \overrightarrow{AB} = a\mathbf{i} + 16\mathbf{j} \qquad \text{and} \qquad \left|\overrightarrow{OB}\right| = 5\sqrt{29}$$

(a) Find the possible values of a

(4)

Given that $a > 0$

(b) find a unit vector that is parallel to $\overrightarrow{AB}$

(2)

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5. Question 4 · part b · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 6 marks)

5. Question 4 · part b · 2 marks · Working and answer

Working and answer

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6. Question 4 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

4 The points A, B, C and D are the vertices of a quadrilateral $ABCD$ such that

$\overrightarrow{AB} = 7\mathbf{i} + p\mathbf{j}$ $\overrightarrow{AC} = 11\mathbf{i} - p\mathbf{j}$ $\overrightarrow{AD} = 4\mathbf{i} - 2p\mathbf{j}$

(a) Show that, for all values of p, $ABCD$ is a parallelogram.

(3)

Given that $|\overrightarrow{BD}| = 3\sqrt{10}$

(b) find the possible values of p.

(3)

Given that $p > 0$

(c) find a unit vector which is parallel to $\overrightarrow{BD}$.

(1)

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6. Question 4 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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6. Question 4 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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6. Question 4 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 7 marks)

6. Question 4 · part c · 1 mark · Working and answer

Working and answer

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7. Question 3 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

3 O, A and B are fixed points such that

$$\overrightarrow{OA} = (p\mathbf{i} - 4\mathbf{j}) \qquad \overrightarrow{OB} = \mathbf{i} + (2p + 1)\mathbf{j}$$

Given that $\sqrt{2}\left|\overrightarrow{OA}\right| = \left|\overrightarrow{OB}\right|$ and $p > 0$

(a) find the value of p

(4)

Using this value of p

(b) find a unit vector that is parallel to $\overrightarrow{AB}$

(5)

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7. Question 3 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 9 marks)

7. Question 3 · part b · 5 marks · Working and answer

Working and answer

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8. Question 3 · part b · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

3 O, A and B are fixed points such that

$$\left|\overrightarrow{OA}\right| = 3\sqrt{5} \qquad \overrightarrow{AB} = \mathbf{i} + 3a\mathbf{j} \qquad \overrightarrow{OB} = 7\mathbf{i} + 2a\mathbf{j}$$

Given that $a > 0$

(a) find the value of a

(4)

(b) Hence find a unit vector parallel to $\overrightarrow{OA}$

(2)

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8. Question 3 · part b · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

8. Question 3 · part b · 2 marks · Working and answer

Working and answer

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9. Question 10 · part whole · 10 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

10 O, A and B are fixed points such that

$\overrightarrow{OA} = (b + 1)\mathbf{i} + b\mathbf{j}$

$\overrightarrow{AB} = 3\mathbf{i}$

The unit vector parallel to $\overrightarrow{OB}$ is $\frac{\sqrt{17}}{34}[(3a + 2)\mathbf{i} + b\mathbf{j}]$

Given that a and b are constants where $a > 0$ and $b > 0$

find the exact value of

(i) a

(ii) b

(10)

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9. Question 10 · part whole · 10 marks

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Further practice · Attempt part whole only.

Question 10 continued

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9. Question 10 · part whole · 10 marks

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Question 10 continued

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9. Question 10 · part whole · 10 marks

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Question 10 continued

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(Total for Question 10 is 10 marks)

9. Question 10 · part whole · 10 marks · Working and answer

Working and answer

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10. Question 4 · part c · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = p\mathbf{i} + 2p\mathbf{j} \qquad \overrightarrow{OB} = 5\mathbf{i} + 9p\mathbf{j}$$

Given that $\overrightarrow{AB}$ is parallel to $(\mathbf{i} - 2\mathbf{j})$

(a) find the value of p

(6)

(b) Hence find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(c) Find a unit vector parallel to $\overrightarrow{OA}$

Give your answer in the form $\frac{\sqrt{a}}{5}(b\mathbf{i} + c\mathbf{j})$ where a, b and c are integers to be found.

(4)

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10. Question 4 · part c · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 12 marks)

10. Question 4 · part c · 4 marks · Working and answer

Working and answer

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11. Question 6 · part b · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 Referred to a fixed origin O, the position vector of the point A is $(9\mathbf{a}+12\mathbf{b})$ and the position vector of the point B is $(2\mathbf{a}+p\mathbf{b})$ where $|\mathbf{a}|=|\mathbf{b}|=1$

Given that $|AB|=25$

(a) show that the two possible values of p are -12 and 36

(4)

For $p=-12$

(b) find a unit vector parallel to $\overrightarrow{OB}$
Give your answer in terms of **a** and **b** in its simplest form.

(2)

Point C lies on OB such that $OC:CB=2:3$

For $p=36$

(c) find $\overrightarrow{CA}$
Give your answer in terms of **a** and **b** in its simplest form.

(3)

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11. Question 6 · part b · 2 marks · Working and answer

Working and answer

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12. Question 9 · part b · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

9 The points A, B, C and D are the vertices of a quadrilateral where

$$\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b} \qquad \overrightarrow{AC} = 7\mathbf{a} + 7\mathbf{b} \qquad \overrightarrow{AD} = 10\mathbf{a} + 6\mathbf{b}$$

(a) Show that $ABCD$ is a trapezium.

(4)

Given that $|\mathbf{a}| = 1$ and $|\mathbf{b}| = 1$ and that $\mathbf{a}$ and $\mathbf{b}$ are perpendicular to each other

(b) find a unit vector parallel to $\overrightarrow{BD}$

(4)

The point Y lies on CD such that $CY : YD = 1 : 2$

The lines AC and BY intersect at the point X

(c) (i) Use a vector method to find the ratio $BX : XY$

(4)

(ii) Hence find, in its simplest form, the ratio

area of triangle CXY : area of trapezium $ABCD$

(4)

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12. Question 9 · part b · 4 marks · Working and answer

Working and answer

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